Valencia Balam Susanto

+852 5972 8389 | valenciabalamm@gmail.com | valenciabalamsusanto.github.io

EDUCATION

The Hong Kong Polytechnic University
BEng (Hons) in Electronic Systems & Internet-of-Things *2023 — 2027*

- Expected **First-Class Honours** | FEng Dean's List | Recipient of PolyU Post Entry Scholarship
- Related Subjects: AI & Data Analytics (A+), Electronic System Training (A-), Machine Learning (A+)

WORK EXPERIENCE

MTR Corporation Ltd Hong Kong
Information Technology & AI Intern *Jul — Aug 2025*

- Led end-to-end development and deployment of **2 internal RAG chatbots**, adopted by ~100 users, raising both enterprise AI adoption and workplace **efficiency by 20%+**.
- Built an **LLM-as-Judge tool in Python** to automate QA for the 2 RAG chatbots, cutting turnaround time and errors by **50%+** and manual testing effort by **30%+**.
- Presented chatbot demos to technical and business stakeholders, including the CIO.

BeeInventor Ltd Hong Kong
Information Engineering Intern *Jun — Aug 2024*

- Analysed large-scale temperature data using **Python, SQL, and Excel**. Successfully identified sensor calibration issues and ~5 helmets with faulty temperature sensors in active use.
- Refined sensor software and embedded C firmware, improving temperature reading **accuracy by 5%+**.
- Collaborated with 3+ senior engineers and the business team on both IoT QA and development.

The Hong Kong Polytechnic University Hong Kong
Part-time Research Student Assistant *Sep 2023 — Present*

- Support Dr. Jeffry Oktavianus (Assistant Professor) **across 5+ research projects** in data collection, dataset management and exploratory AI research.
- Collect and clean research data in Excel and Python, **delivering 3+ structured, analysis-ready datasets** with summaries of key insights and trends.

PROJECTS

Large Language Models | *Applying LLMs for Bahasa Indonesia Scam Detection (Final Year Project)*

- Building an automated Python experiment pipeline that batch-processes 200+ scam call transcripts, replacing manual prompting to **compare 3 LLMs and RAG setups** quickly and repeatably.

Machine Learning | *Automated Loan Approvals & Default Prediction*

- Built approval and default-risk models in Python on 1.6M Lending Club applications and 546K loans, using XGBoost, scorecards and SHAP. Trained a machine learning model that out-ranked Lending Club's own grades on later loans (**Gini 0.385 vs 0.357**).

Exploratory Data Analysis | *Sales & Market Analysis of Electronic Products on Amazon*

- Cleaned 42,675 Amazon listings into 8,149 electronics products for pricing, discounts, trends and market basket analysis. **Built an interactive dashboard** to consolidate findings and guide seller decisions.

SKILLS

- **Languages:** English, Indonesian, Chinese
- **Programming:** Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn), SQL, C, C++, HTML, CSS, Javascript
- **Data & AI:** Excel, Power BI, Tableau, Predictive Modelling, RAG, LLM Applications & Evaluation

LEADERSHIP & VOLUNTEERING

- **Coordinator** | *STEM & AI Outreach (Philippines)* *Jan 2026*
  Coordinated logistics and lesson delivery surrounding AI and object detection for 15 children.
- **Head of Media** | *Indonesian Student Association (PolyU)* *Sep 2024 — Aug 2025*
  Led a team of 2 to plan and produce both social media content and event publicity for 6+ events.
- **English Teacher** | *IMWU* *Sep 2023 — Sep 2024*
  Taught English to 50+ Indonesian migrant domestic workers in Hong Kong.